

Beyond Regularity:

Stronger Revenue Guarantees with MHR Distributions in Testable
Learning

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Abstract

This thesis extends the Testable Learning Framework for auction mechanism design to distributions with Monotone Hazard Rate (MHR) properties, providing both theoretical guarantees and empirical validation. While optimal auction design traditionally relies on precise knowledge of bidders' value distributions, real-world scenarios involve uncertainty and potential irregularity in these distributions. The Testable Learning Framework addresses this challenge by pairing a tester with a learner: the tester verifies when near-optimal auctions can be learned from available data, while the learner constructs such auctions when possible. I demonstrate that MHR distributions—where the probability of a value occurring, conditional on being at least that high, increases with value—permit tighter revenue guarantees than regular distributions. Specifically, MHR distributions achieve approximation factors of $(1 - O(n\alpha))$ versus $(1 - O(\sqrt{n\alpha}))$ for regular distributions, where α represents the distortion parameter and n the number of bidders. This improvement is significant for practical applications where α is small.

Through empirical testing on synthetic data, I validate the theoretical predictions and examine the framework's robustness to distributional misspecification. The tester behaves largely as expected, accepting when $\alpha \geq \varepsilon$ and rejecting otherwise, with learned revenue approaching theoretical optimality as α decreases. I also analyze potential improvements to sample complexity bounds by identifying the sources of logarithmic factors in current bounds and suggesting possible refinements.

This research contributes to bridging the gap between theoretical auction design and practical implementation by providing mechanisms that perform well without perfect knowledge of bidders' distributions. The extended framework enables auction designers to confidently deploy mechanisms with strong revenue guarantees, even when facing uncertain or irregular value distributions.

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1 Introduction

Since their inception, spectrum auctions by the U.S. Federal Communications Commission generated over \$223 billion in revenue (Lee and Malamud (2023)). Similarly, sponsored search results on major search engines rely on auction mechanisms. These high-stakes auctions face a critical challenge: uncertainty about bidders’ value distributions. When small misspecifications occur, significant revenue losses can follow. Imagine a telecommunications company valuing spectrum rights at \$5 billion while the auction mechanism, operating on incorrect distribution assumptions, sets a reserve price of \$7 billion, resulting in no sale and zero revenue. This scenario highlights a fundamental question in mechanism design: how can we ensure near-optimal revenue without perfect knowledge of bidders’ value distributions?

Classical auction theory, beginning with Myerson’s work, offers elegant solutions for revenue maximization when bidders’ values follow regular distributions. Regularity requires a monotonically increasing virtual value function—a mathematically convenient but often unrealistic assumption. Real-world auction data frequently violates this condition. More fundamentally, with limited historical data, how can auction designers determine whether their mechanism will perform well when regularity may not hold?

The Testable Learning Framework (TLF) addresses this challenge by evaluating a mechanism’s near-optimality using only a small sample of bids. TLF resolves a fundamental issue: with insufficient samples, we cannot perfectly learn the optimal mechanism, yet decisions must be made. The framework provides a binary verdict and accepts mechanisms with strong revenue guarantees under regularity assumptions and rejects those without.

While TLF represents significant progress, several critical questions remain unanswered. How does the framework perform when distributions meet stricter criteria, such as the Monotone Hazard Rate (MHR) property? MHR distributions—where the probability of a value occurring, conditional on being at least that high, increases with the value—are both common in practice and theoretically significant. They permit improved revenue guarantees

and potentially more efficient testing procedures. Yet we lack systematic understanding of TLF’s performance on such distributions or how to adapt the framework to leverage their additional structure.

This thesis extends the Testable Learning Framework to MHR distributions, both theoretically and empirically. We investigate three central questions:

1. How do TLF’s theoretical guarantees change when assumptions strengthen from regularity to MHR?
2. In practice, how sensitive is TLF to slight violations of these distributional assumptions?
3. Can we achieve improved sample complexity bounds by exploiting the MHR structure?

We demonstrate that TLF can be refined to provide stronger guarantees for MHR distributions compared to regular distributions. We also develop and validate empirical tests on synthetic data to identify how frequently TLF accepts or rejects in real-world scenarios and how learned revenue compares to theoretical optimals.

The contributions of this thesis are three-fold: (1) new theoretical bounds on the revenue gap between learned and optimal mechanisms when bidder values follow MHR distributions; (2) empirical tests evaluating TLF’s robustness to various forms of distributional misspecification; and (3) identification of potential improvements to the sample complexity of auctions.

The remainder of this thesis proceeds as follows. Chapter 2 reviews necessary background in auction theory and prior work on TLF. Chapter 3 introduces core concepts of TLF as applied to auctions and develops the theoretical foundation for its extension to MHR distributions. Chapter 4 presents empirical findings on the framework’s accept/reject decisions and analyzes revenue gaps between learned and optimal mechanisms. It also extends TLF to MHR distributions, including theoretical guarantees and experimental validation, while exploring sample complexity improvements. We conclude with implications for practical auction design and directions for future research.

2 Prior Results

2.1 Introduction to Auction Theory

Auctions allocate resources throughout the economy, with their design significantly impacting efficiency and revenue. Auction designers face a central challenge: creating rules that incentivize truthful revelation of private information while maximizing revenue.

Consider three bidders competing for a painting in a sealed-bid auction. In a first-price auction, each bidder submits a private bid, and the highest bidder wins, paying exactly their bid amount. This format creates strategic complexity—bidders have incentives to bid below their true valuations. As noted in Roughgarden (2016), you should never bid your valuation in a first-price auction (that would guarantee zero utility), and the ideal amount to underbid depends on the other players' bids.

A second-price (Vickrey) auction resolves this dilemma: the highest bidder wins but pays the second-highest bid. This mechanism makes truthful bidding a dominant strategy. A bidder with a \$100 valuation gains nothing by bidding differently—bidding above \$100 risks paying more than her value, while bidding below only reduces her chances of winning without affecting the payment when she does win.

2.1.1 Key Auction Theory Concepts

Expected Revenue is the average payment the auctioneer expects to receive from the bidders. For a mechanism with allocation rule x and payment rule p , the expected revenue is:

$$\text{Rev}(M, D) = \mathbb{E}_{v \sim D} \left[\sum_{i=1}^n p_i(v) \right]$$

where $v = (v_1, v_2, \dots, v_n)$ represents the bidders' valuations drawn from distribution D , and $p_i(v)$ is the payment made by bidder i .

Social Welfare (or simply welfare) represents the total value generated by the auction. It

is the sum of the values that each bidder places on their allocation:

$$\text{Welfare}(M, D) = \mathbb{E}_{v \sim D} \left[\sum_{i=1}^n v_i \cdot x_i(v) \right]$$

where $x_i(v)$ indicates the probability that bidder i receives the item. In single-item auctions, welfare is maximized by allocating the item to the bidder with the highest valuation.

Dominant Strategy Incentive Compatibility (DSIC) means that truthful bidding is optimal for each bidder regardless of others' bids. A mechanism is DSIC if:

$$u_i(v_i, \vec{\mathbf{b}}_{-i}) \geq u_i(b_i, \vec{\mathbf{b}}_{-i}) \quad \forall \quad i \in [n], \quad v_i \in \mathbb{R}_{\geq 0}, \quad b_i \in \mathbb{R}_{\geq 0}, \quad \mathbf{b}_{-i} \in \mathbb{R}_{\geq 0}^{n-1}$$

where $u_i(b_i, \vec{\mathbf{b}}_{-i}) = v_i \cdot x_i(b_i, \vec{\mathbf{b}}_{-i}) - p_i(b_i, \vec{\mathbf{b}}_{-i})$ is bidder i 's utility when they bid b_i while having true value v_i , and other bidders bid $\vec{\mathbf{b}}_{-i}$. Each bidder's utility is maximized when bidding exactly their true valuation: $b_i = v_i$.

Individual Rationality (IR) ensures that bidders never experience negative utility when participating truthfully. Formally, a mechanism satisfies IR if:

$$u_i(v_i, \vec{\mathbf{b}}_{-i}) \geq 0 \quad \forall \quad i \in [n], \quad v_i \in \mathbb{R}_{\geq 0}, \quad \mathbf{b}_{-i} \in \mathbb{R}_{\geq 0}^{n-1}$$

For revenue-optimal auctions, Myerson introduced **virtual valuations**:

$$\phi(v) = v - \frac{1 - F(v)}{f(v)}$$

where v is the bidder's value, F is the cumulative distribution function, and f is the probability density function. Myerson's breakthrough finding was that an auction's expected revenue equals the expected virtual welfare—the virtual value of the winner. This insight characterized revenue-optimal auctions through virtual values.

Myerson proved that the revenue-optimal mechanism allocates the item to the bidder

with the highest non-negative virtual valuation (Myerson (1981)). However, this assumes virtual valuations increase monotonically, which is a condition known as regularity. While regularity has become central to auction theory, many real-world distributions violate this assumption.

2.2 Cole and Roughgarden (2014)

Cole and Roughgarden explored how many samples from bidders' value distributions are needed to learn revenue-maximizing auctions when distributions aren't known a priori.

They framed this as a learning task with an approximation benchmark, seeking a mechanism that achieves a $(1 - \varepsilon)$ fraction of optimal revenue with high probability. Their key finding is that $m = \text{poly}(k, \frac{1}{\varepsilon}) = \Omega(\frac{k^{10}}{\varepsilon^7} \ln^3 \frac{k}{\varepsilon})$ samples are necessary and sufficient to generate an auction strategy for k bidders that achieves expected revenue at least $(1 - \varepsilon)$ times that of the optimal auction.

2.3 Guo, Huang, and Zhang (2019)

Guo, Huang, and Zhang built on Cole and Roughgarden's work to definitively settle the sample complexity question. They developed an analytical technique combining strong revenue monotonicity and information theory to find tighter sample complexity bounds that match their corresponding lower bounds up to poly-logarithmic factors.

Their findings effectively closed the theoretical question of sample complexity for single-parameter revenue maximization, establishing the following bounds:

- **Regular Distributions:** $m = O(n\varepsilon^{-3} \ln(n\varepsilon^{-1}) \ln(n\delta^{-1}\varepsilon^{-1})) = \tilde{O}(n\varepsilon^{-3})$
- **MHR Distributions:** $m = O(n\varepsilon^{-2} \ln(\varepsilon^{-1}) \ln(n\varepsilon^{-1}) \ln(n\delta^{-1}\varepsilon^{-1})) = \tilde{O}(n\varepsilon^{-2})$

These sample sizes guarantee expected revenue at least $(1 - \varepsilon) \cdot \text{OPT}(D)$ with probability at least $1 - \delta$.

2.4 Guo, Jordan, and Zampetakis (2021)

While earlier work assumed perfect samples, Guo, Jordan, and Zampetakis tackled the more realistic scenario of corrupted or adversarially perturbed distributions. They introduced algorithms that perform well even when training on “approximate distributions” differing from true distributions in Kolmogorov-Smirnov distance. Their key innovation was the minimal regular (or MHR) distribution—a distribution within a specified Kolmogorov ball that is stochastically dominated by every other distribution in the ball.

A distribution $\mathcal{D}_1$ is said to stochastically dominate another distribution $\mathcal{D}_2$ (denoted as $\mathcal{D}_1 \succeq \mathcal{D}_2$) if for every value x , the probability that a sample from $\mathcal{D}_1$ exceeds x is at least as large as the corresponding probability for $\mathcal{D}_2$. Formally, $\mathcal{D}_1$ stochastically dominates $\mathcal{D}_2$ if $F_{\mathcal{D}_1}(x) \leq F_{\mathcal{D}_2}(x)$ for all x , where $F_{\mathcal{D}}$ represents the cumulative distribution function. A product distribution $\mathbf{D} = \Pi_i \mathcal{D}_i$ (component-wise) stochastically dominates another product distribution $\mathbf{D}' = \Pi_i \mathcal{D}'_i$ if $\mathcal{D}_i \succeq \mathcal{D}'_i \quad \forall i$.

This property is crucial because stochastic dominance preserves certain ordering relationships between mechanisms’ revenues. This property, in addition to *strong revenue monotonicity*, was used to show that generating Myerson’s empirically optimal mechanism on this minimal distribution still produces good revenue.

Strong revenue monotonicity states that for two product distributions $\mathbf{D}, \mathbf{D}'$ such that $\mathbf{D}' \succeq \mathbf{D}$, the optimal mechanism M for $\mathbf{D}$ adheres to the following property:

$$\text{Rev}(M, \mathbf{D}) \leq \text{Rev}(M, \mathbf{D}')$$

For MHR distributions, they proved a $(1 - O(\alpha))$ approximation ratio, while regular distributions achieve a $(1 - O(\sqrt{\alpha}))$ ratio, where α represents the Kolmogorov-Smirnov distance between the true and corrupted distributions. This parameter α quantifies the maximum pointwise difference between the cumulative distribution functions of the true and corrupted

distributions. They also established these bounds as tight by providing matching lower bounds.

By accounting for data corruption, the authors expanded practical applicability of learning-based auction design and demonstrated how MHR and regular distributions have different performance guarantees.

2.5 Safely Learning Optimal Auctions: A Testable Learning Framework for Mechanism Design (Working Paper)

The most recent advance comes from a working paper introducing the Testable Learning Framework (TLF) for mechanism design. This work bridges the gap between theoretical guarantees that rely on distributional assumptions and practical applications where those assumptions may not hold.

Unlike Guo et al. (2021), which allowed for corrupted input data, this paper develops an algorithm that handles *irregular* inputs. The authors present a paired tester and learner: the tester verifies whether distributional assumptions are satisfied from the given samples.

Formally, the framework defines what it means for a tester $\mathcal{T}$ to be a regularized tester with respect to a learner $\mathcal{L}$. Let $\mathbf{D}$ be any product distribution, n be the number of samples, and $\alpha \in (0, 1)$ be a distortion parameter. The tester $\mathcal{T}$ must satisfy:

- **Soundness:** If $\mathcal{T}$ accepts a distribution $\mathbf{D}$ with probability at least $1 - \delta$, then with probability at least $1 - \delta$, the learner $\mathcal{L}$ outputs a mechanism M such that:

$$\text{Rev}(M, \mathbf{D}) \geq (1 - g(\alpha, n))\text{OPT}(\mathbf{D}')$$

for all regular distributions $\mathbf{D}'$ that are α -close to $\mathbf{D}$ in Kolmogorov-Smirnov distance.

- **Completeness:** If $\mathbf{D}$ is a regular distribution, then $\mathcal{T}$ accepts $\mathbf{D}$ with probability at least $1 - \delta$.

Here, $g(n, \alpha)$ represents the approximation factor, which is $O(\sqrt{n\alpha})$ for regular distributions and we show can be improved to $O(n\alpha)$ for MHR distributions. When the tester accepts, it guarantees the learning algorithm will achieve good performance relative to a nearby regular distribution. When it rejects, it certifies that the distribution violates regularity.

This framework allows auction designers to confidently deploy mechanisms even without certainty about the regularity of bidder distributions. When the tester accepts, the auction achieves revenue at least $(1 - O(\sqrt{n\alpha})) \cdot \text{OPT}(\mathbf{D}')$ for every regular distribution $\mathbf{D}'$ that is α -close to the input distribution, where n is the number of bidders.

3 The Testable Learning Framework for Mechanism Design

3.1 The Challenge: Distributional Assumptions in Auction Theory

Optimal auction design depends critically on knowing bidders' value distributions. In practice, auction designers rarely have this perfect knowledge, instead working with limited samples of past bids. This creates a fundamental challenge: how do we design auctions that perform well with incomplete distributional information?

Theoretical guarantees in auction theory typically assume bidders' value distributions are regular or satisfy the stronger MHR property. Real-world data often violates these assumptions, creating a gap between theory and practice.

3.2 Understanding Regularity and MHR Distributions

A distribution is **regular** when its virtual value function increases monotonically:

$$\phi(v) = v - \frac{1 - F(v)}{f(v)}$$

While mathematically precise, this definition lacks intuition. Regular distributions include many common distributions such as uniform, exponential, and normal distributions.

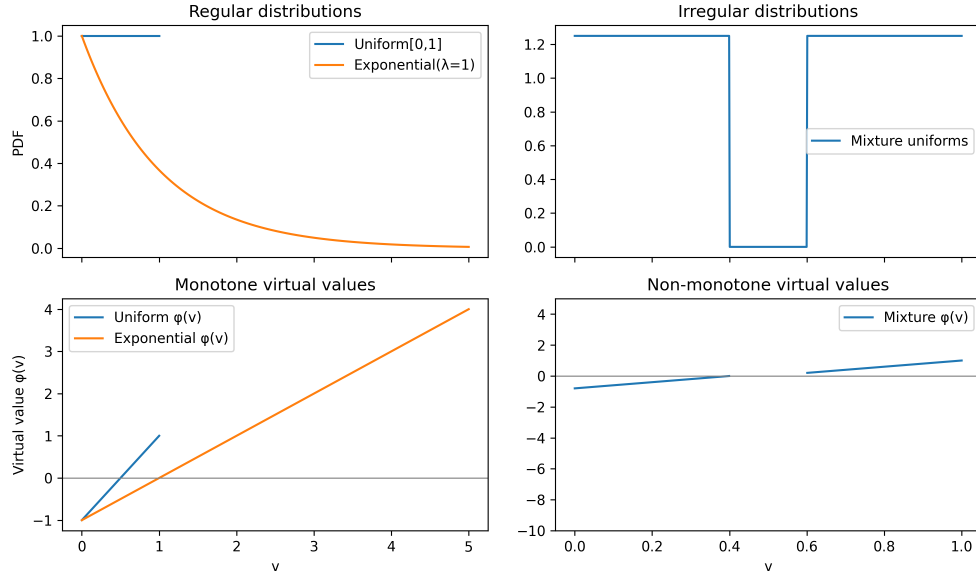


Figure 1: Regular vs. Irregular Distributions

Figure 1 illustrates the difference: uniform and exponential distributions have monotonically increasing virtual values, while the mixed distribution has a sometimes non-differentiable virtual value function, making it irregular. Regularity ensures that the optimal auction has a simple structure where higher bids correspond to higher chances of winning.

A **monotone hazard rate (MHR)** distribution satisfies an even stronger condition. Its hazard rate function $h(v) = \frac{f(v)}{1-F(v)}$ must increase monotonically. Intuitively, this means the probability of a value occurring, given that it's at least that high, increases with the value.

MHR distributions form a strict subset of regular distributions: every MHR distribution is regular, but not every regular distribution is MHR. Examples include uniform, exponential, and normal distributions with positive support.

These properties matter because Myerson's optimal auction allocates items based on

virtual values. When virtual values aren’t monotonic (irregular distributions), the optimal auction requires an “ironing” procedure because empirical distributions are generally irregular, even when the underlying distributions are regular. This procedure can be difficult with small samples because the virtual value function can change arbitrarily rapidly in regions where density changes rapidly (Cole and Roughgarden (2015)).

3.3 The Core Insight of Testable Learning

The Testable Learning Framework (TLF) addresses this challenge with a novel approach. Rather than attempting to directly test whether a distribution is regular—which is provably impossible with finite samples—TLF introduces two components:

1. A **tester** that determines whether a near-optimal auction can be learned from available samples
2. A **learner** that constructs such an auction when the test passes

This shifts focus from the abstract question “Is this distribution regular?” to the more practical question “Can we learn a near-optimal auction from this data?”

3.4 The Impossibility of Direct Testing

Directly testing for regularity with finite samples is impossible. Consider two distributions: one perfectly regular, and another irregular but with the irregularity concentrated in a tiny region of extremely low probability.

No finite sample can reliably distinguish between these distributions because the samples that would reveal the irregularity occur with vanishingly small probability. Yet this tiny irregularity can drastically change the optimal auction design and achievable revenue. We can’t hope to achieve the optimal revenue because we have two statistically indistinguishable distributions (with any practical sample size) that require fundamentally different auction mechanisms to maximize revenue.

For example, consider a uniform distribution on $[0, 1]$ versus a mixture distribution that is uniform on $[0, 1]$ with probability $(1 - 10^{-9})$ and uniform on $[10^9, 10^9 + 1]$ with probability 10^{-9} . For the simple uniform distribution, the optimal auction sets a reserve price of 0.5 and achieves an expected revenue of 0.25. For the mixture distribution, however, the optimal mechanism sets a reserve price of approximately 10^9 , achieving an expected revenue of 1 ($10^9 \times 10^{-9} = 1$), which is four times higher than for the pure uniform distribution.

Yet to reliably detect this difference and implement the correct mechanism, we would need approximately 10^9 samples—a practical impossibility in most real-world scenarios. This challenge highlights why we can’t directly test for regularity or optimality with limited samples.

This result motivates TLF’s innovation: instead of testing the distributional property directly, we test whether the data supports learning a near-optimal auction.

3.5 Regularized Testable Learning

The regularized version of TLF introduces a crucial modification. When the tester accepts, it guarantees the learned auction achieves revenue that is a certain fraction of the optimal revenue of any regular distribution α -close to the input distribution.

This guarantee, while weaker than traditional learning guarantees, is more realistic. It ensures either the distribution is close to regular, in which case our learned auction performs well, or it’s far from regular, in which case the test fails.

The testing procedure uses the *link function* of distributions. For regular distributions, this function is:

$$h_r(x; F) = \frac{1}{1 - F(x)}$$

A distribution is regular if and only if its link function is convex. The tester estimates this link function from samples and checks whether its convex envelope (the smallest convex

function below it) is close to the original function.

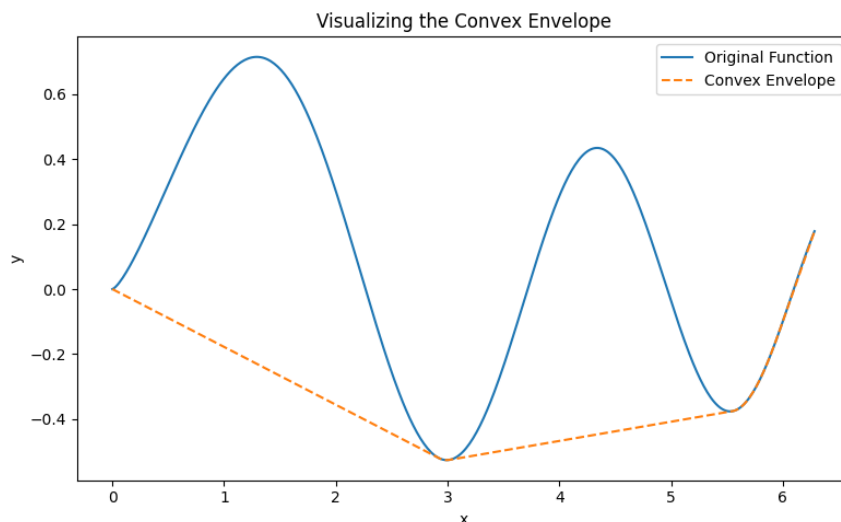


Figure 2: Illustration of the Convex Envelope

3.6 The TLF Algorithms

The TLF consists of two key algorithms:

3.6.1 Regularized Tester

The tester determines whether a near-optimal auction can be learned from given samples:

1. Construct an empirical distribution from the samples
2. Create a dominated version by shifting quantiles
3. Make this dominated distribution regular by taking the convex envelope of its link function
4. Check if this regular approximation is close to the original empirical distribution
5. Accept if yes; reject otherwise

3.6.2 Regularized Learner

When the tester accepts, the learner constructs an auction:

1. Follow the same initial steps as the tester
2. Create a further conservative version through additional quantile shifting
3. Output Myerson’s optimal auction for this conservative distribution

This conservative approach ensures the learned auction performs well not just on the empirical distribution but on any nearby regular distribution.

3.7 Revenue Guarantees and Their Dependence on Distribution Class

TLF’s revenue guarantees depend on the distribution class:

- For **regular distributions**, the guarantee is $(1 - O(\sqrt{n\alpha}))$ of optimal revenue
- For **MHR distributions**, we expect a tighter guarantee of $(1 - O(n\alpha))$

This difference is significant. With a distortion parameter $\alpha = 0.01$ and a single bidder ($n = 1$), the MHR guarantee promises approximately 99% of optimal revenue, while the regular guarantee offers only about 90%.

The distinction arises because MHR distributions have stronger concentration properties, allowing for more precise revenue estimations.

3.8 Applications Beyond Revenue Maximization

The TLF approach extends beyond simple revenue maximization to test the applicability of fundamental auction theory results:

1. **Bulow-Klemperer Theorem:** Is adding one more bidder better than optimizing reserve prices? This theorem states that when bidders’ values are drawn from i.i.d. regular distributions, the revenue of a simple Vickrey auction with one additional bidder exceeds the revenue of an optimal mechanism with one fewer bidder. In other words, competition is more important than optimality in i.i.d. settings. Applying the TLF, when the tester approves an input distribution, it also holds that the Vickrey

an auction with an additional bidder achieves at least $(1 - O(\sqrt{n\alpha}))$ of the optimal revenue for any regular distribution α -close to the input distribution. Formally,

$$\text{Rev}_{n+1}(VA, D) \geq (1 - O(\sqrt{n\alpha})) \cdot \text{OPT}_n(D')$$

where D' is an α -close regular distribution.

2. Anonymous Price Auctions: Can a single reserve price achieve good revenue?

When bidders have non-identical regular distributions, a well-known result by Hartline and Roughgarden (2009) states that an auction with a single anonymous reserve price (the same reserve price for all bidders) can achieve a constant fraction of the optimal revenue of Myerson's auction. The TLF tester determines that this simplification applies, and when it accepts, the anonymous price auction achieves at least $\frac{1}{e}(1 - O(\sqrt{n\alpha}))$ of the optimal revenue for any regular distribution D' that is α -close to the input distribution. Formally,

$$\text{Rev}(\text{APA}(p^*), \mathbf{D}) \geq \frac{1}{e}(1 - O(\sqrt{n\alpha}))\text{OPT}'(\mathbf{D})$$

where $\mathbf{D} = \Pi_{i=1}^n D_i$ and $\mathbf{D}' = \Pi_{i=1}^n D'_i$ are a product distributions where D_i is regular for all i and D'_i is α -close to D_i (within the Kolmogorov ball).

4 Findings

4.1 Empirically Testing the Testable Learning Framework

We implemented the tester-learner pair in Python and evaluated its performance across different input types and distributions.

4.1.1 Evaluating Tester Acceptance Frequency

To assess the tester’s sensitivity, we constructed examples varying the input parameters α and ε . Our primary test distribution drew samples from $U[0, 1]$ with probability $(1 - \varepsilon)$ and from $U[2^{1/\varepsilon}, 2^{1/\varepsilon} + 1]$ with probability ε .

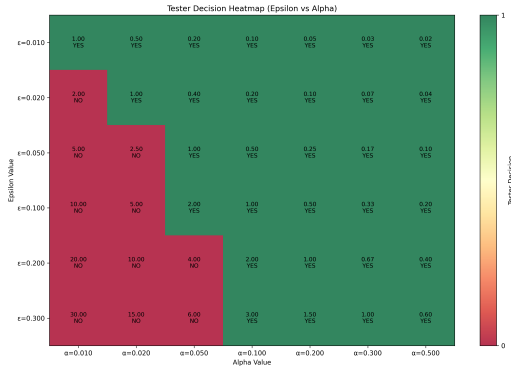


Figure 3: Tester Decision Heatmap, Dynamic m



Figure 4: Tester Decision Heatmap, Fixed and Large m

In both figures, we tested $\alpha = [0.01, 0.02, 0.05, 0.1, 0.3, 0.5]$ and $\varepsilon = [0.01, 0.02, 0.05, 0.1, 0.2, 0.3]$.

Figure 3 uses the sample complexity bound from the paper with a buffer:

$$m = \log\left(\frac{1}{\delta}\right) / \alpha^2 + 100$$

Figure 4 uses a fixed large sample size $m = 500,000$. Theoretically, we would expect the tester to accept when $\alpha \geq \varepsilon$, as the irregular distribution should be at most ε -far from the regular $U[0, 1]$ distribution.

While we see some expected pattern in the first plot, the second plot with larger sample size shows results much closer to the theoretical expectationnearly perfect alignment along the diagonal with only one notable outlier at $(\alpha = 0.02, \varepsilon = 0.03)$. This discrepancy may arise because both the support and component probabilities depend on ε rather than α .

The tester behaves as expected: for a fixed ε , it initially rejects with very small α values (less than ε). This makes sense because regular distributions aren't dense, so finding a close regular distribution is impossible when α is too small. As α increases past ε , the tester begins accepting and continues to accept as α increases further.

We also tested a mixture of normal distributions:

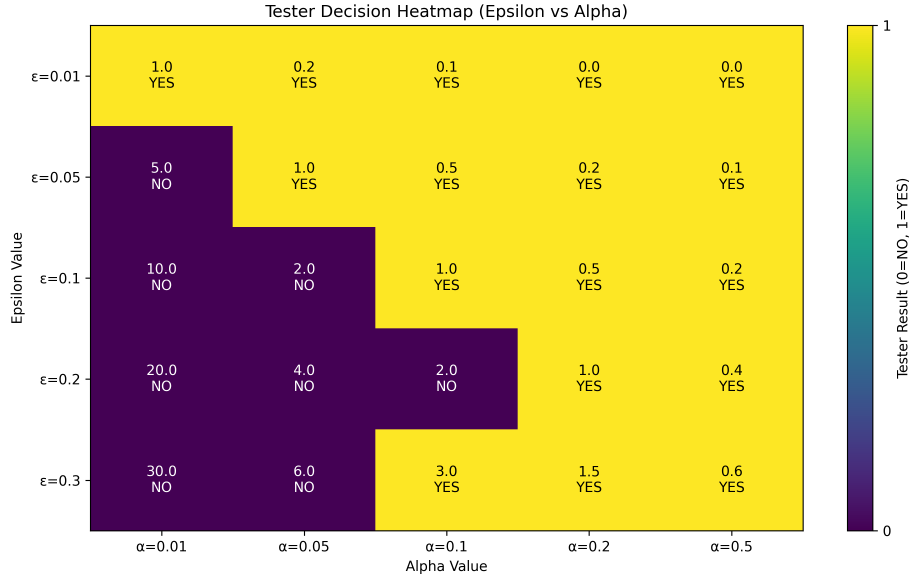


Figure 5: Tester Decision Heatmap, Fixed m for Mixed Normal Distributions

Figure 5 shows results from a mixed distribution of $\mathcal{N}(0, 1)$ with probability $(1 - \varepsilon)$ and $\mathcal{N}(2^{1/\varepsilon}, 1)$ with probability ε , using $m = 500,000$. We again see largely monotonic results along the diagonal, with deviations at $\varepsilon = 0.3$ where the tester accepts for $\alpha = 0.1, 0.2$. This may occur because higher ε values shift probability mass in ways that change the distribution's regularity properties.

We conducted a similar test on a mixture of log-normal distributions, as log-normal

distributions are commonly seen in the literature as a standard assumption for bidder distributions (along with the uniform). The figure below depicts a mixture of two log-normal distributions, the first with mean $\mu = 1$ and standard deviation $\sigma^2 = 1$ and the second with $\mu = \log(2^{1/\varepsilon})$ and $\sigma^2 = 1$.

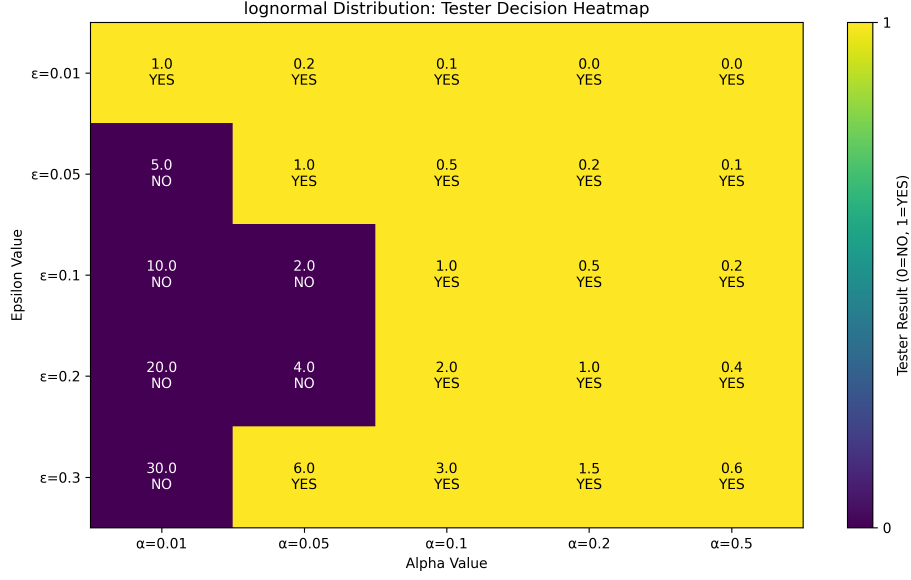


Figure 6: Tester Decision Heatmap, Fixed m for Mixed Log-Normal Distributions

Interestingly, Figure 6 shows a similar trend to that of the mixture of normal distributions with a break in the monotonicity at $\varepsilon = 0.3$. At this value, 30% of the probability mass is at the second distribution (holds for all the distributions). However, due to the higher value of epsilon, the actual location of that probability mass moves closer to the first (by construction of $2^{1/\varepsilon}$ as the location parameter). The effects of both concentrating more probability mass at a closer location may lead to the tester (and learner) finding a *closer* regular distribution than the first distribution, possibly leading to the additional tester acceptances. Further research may explore the specific details as to why the results deviate from expectation at these values.

4.1.2 Comparison of Learned Revenue to Theoretical Optimal

A key aspect of TLF is that its theoretical revenue guarantees are relative to a *nearby regular distribution* rather than the original input. We examined how close the algorithm’s learned revenue came to the theoretical optimal of the actual input distribution.

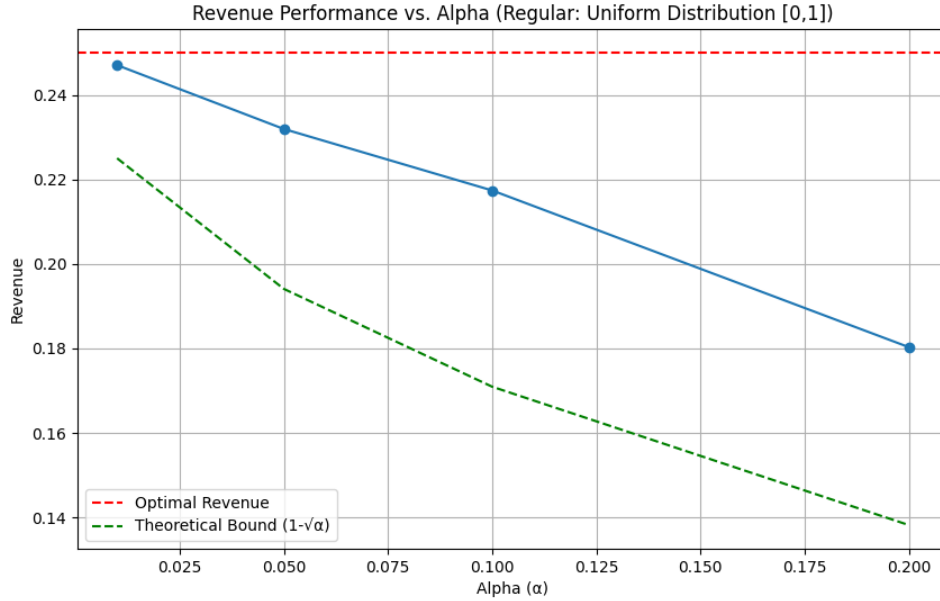


Figure 7: Learned Revenue for $U[0,1]$ (Regular)

Figure 7 shows that for the regular $U[0,1]$ distribution, the algorithm behaves as expected, generating better revenue as α decreases. The optimal strategy sets a reserve price of 0.5, yielding an expected revenue of 0.25. Our learner approaches this theoretical optimum as α decreases. The dotted green line shows the theoretical bound of $O(1 - \sqrt{\alpha}) \cdot OPT$, and our revenue curve follows a similar slope.

Results change with more irregular distributions:

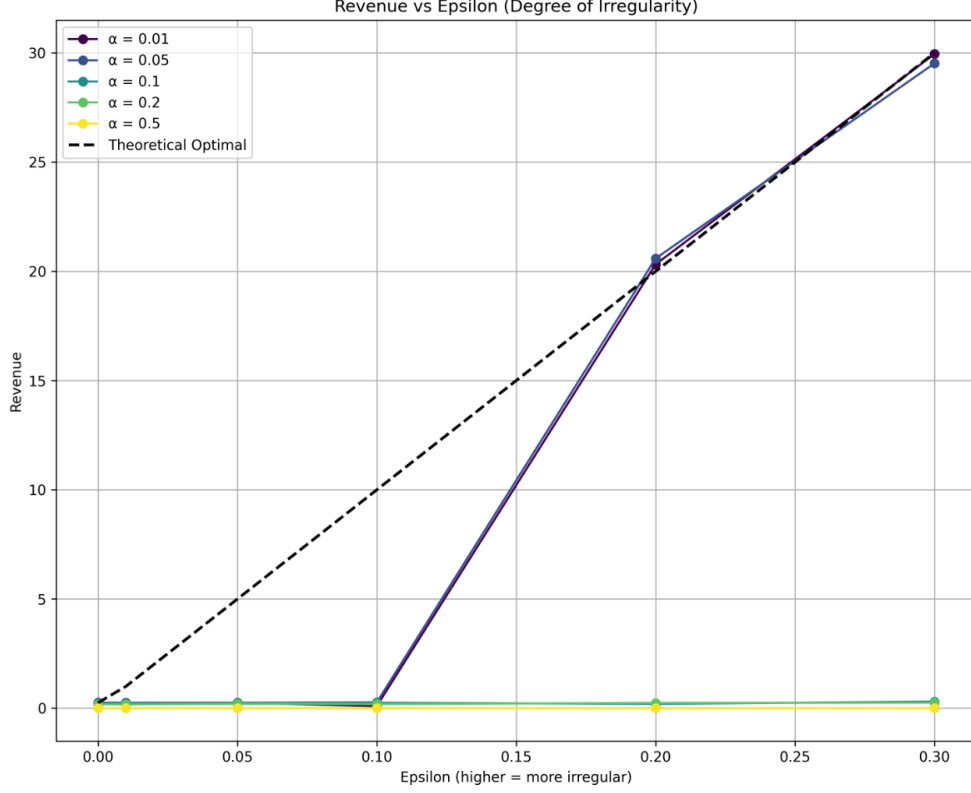


Figure 8: Revenues for Distributions of Varying Irregularity

Figure 8 shows learned revenue for a mixed distribution drawing from $U[0, 1]$ with probability $(1 - \varepsilon)$ and $U[100, 101]$ with probability ε . For large α values, the learner appears to identify $U[0, 1]$ and achieves near its optimal revenue (~ 0.25). Interestingly, for $\varepsilon \in [0.2, 0.3]$ and $\alpha \in [0.01, 0.05]$, the learner somehow identifies the actual optimal revenue for the mixed distribution.

In our testing, these results did not hold for the standard mixed distribution of the uniform over $[0, 1]$ with probability $(1 - \varepsilon)$ and the uniform over $[2^{1/\varepsilon}, 2^{1/\varepsilon} + 1]$ with probability ε . For those distributions, at higher values of epsilon, the learner learned the low revenue of the first uniform distribution and not the optimal revenue. This again may be due to how both the support and concentration of probability are dependent on epsilon by construction, as we saw for tester acceptance.

4.1.3 Directions for Future Research

Our empirical investigation highlights several promising directions for future research:

1. **Multi-parameter dynamics:** Extending this framework to analyze how the tester-learner pair performs with multiple strategic bidders (and multiple items) would provide valuable insights into more complex auction scenarios. Particularly interesting would be examining how revenue guarantees degrade as the number of bidders increases, and whether the $O(\sqrt{n\alpha})$ theoretical bound is tight in practice.
2. **MHR distributions:** While we focused primarily on regular distributions, extending the empirical analysis to MHR distributions could validate the tighter theoretical guarantees of $O(n\alpha)$ versus $O(\sqrt{n\alpha})$. This could lead to more efficient testing procedures specifically tailored to MHR distributions, potentially requiring fewer samples.
3. **Sample complexity optimization:** Our results suggest that in some cases, the theoretical sample complexity bounds may be too conservative. Developing tighter empirical distribution-specific sample complexity bounds and adaptive sampling techniques could improve the efficiency of the tester-learner framework.
4. **Practical implementation:** Developing more methods for constructing the dominated empirical distribution, perhaps using advanced statistical estimation techniques like kernel density estimation or Bayesian methods, could improve performance on real-world bid data.
5. **Real-world validation:** Testing the framework on actual auction data from platforms like eBay or sponsored search auctions would provide insights into its practical applicability and limitations in environments with non-i.i.d. bidders.

4.2 Extension to MHR Distributions

In this section, we extend the findings of the testable learning framework from regular distributions to Monotone Hazard Rate (MHR) distributions. Since the MHR condition is stronger

than regularity (and implies regularity), we know that the findings for regular distributions necessarily hold for MHR distributions, as all MHR distributions are regular. However, we hope to obtain tighter revenue guarantees by eliminating the square root dependency on the distortion parameter α as done in Guo et al. (2021).

4.2.1 Definitions for MHR Distributions

We begin by defining MHR distributions and their associated link function.

Definition 1 (MHR Distribution). A distribution with CDF F and PDF f has Monotone Hazard Rate (MHR) if its hazard rate function $h(v) = \frac{f(v)}{1-F(v)}$ is monotonically non-decreasing for all $v \geq 0$. A product distribution $\mathbf{D}$ is MHR if D_i is MHR for all $i \in [n]$.

Definition 2 (MHR Link Function). For an MHR distribution with CDF F , the link function is defined as:

$$h_M(x; F) = -\ln(1 - F(x)) \quad (1)$$

The corresponding inverse link function is:

$$h_M^{-1}(x; h) = 1 - e^{-h(x)} \quad (2)$$

Lemma 3 (Lemma 3.4 of Guo et al. (2021)). *A distribution D is MHR if and only if its link function $h_M(x; F)$ is a convex function of x . Moreover, for two distributions D and D' , if D first-order stochastically dominates D' then $h_M(x; F) \leq h_M(x; F')$ for all $x \geq 0$.*

4.2.2 Algorithm Adjustments for MHR Distributions

We adapt Algorithms 1 and 2 from the original framework to use the MHR link function. The primary modification is in line 7 of both algorithms, where the regular link function is replaced with the MHR link function.

Algorithm 1 MHR Regularized Tester

Require: m i.i.d. samples from a distribution $\mathcal{D}$ and a distortion parameter α

Ensure: YES or NO

- 1: Let $E = \prod_{i=1}^n E_i$ be the empirical distribution over the samples
 - 2: **for** $i \leftarrow 1$ to n **do**
 - 3: Construct $\hat{E}_i$ as follows:
 - 4: Let $q_{E_i}(v)$ be the quantile function of E_i
 - 5: Define

$$c = \sqrt{\frac{2q_{E_i}(v)(1-q_{E_i}(v)) \ln(2mn\delta^{-1})}{m}} + \frac{4 \ln(2mn\delta^{-1})}{m} + \alpha$$
 - 6:
$$q_{\hat{E}_i}(v) = \begin{cases} \max(0, q_{E_i}(v) - c) & \text{if } v > 0, \\ 1 & \text{if } v = 0 \end{cases}$$
 - 7: $\tilde{E}_i = h_M^{-1}(\text{Conv}(h_M(\hat{E}_i(\cdot))))$
 - 8: **for** each value v in the support of E_i **do**
 - 9: **if** $|q_{\tilde{E}_i}(v) - q_{E_i}(v)| > c$ **then**
 - 10: **return** NO
 - 11: **end if**
 - 12: **end for**
 - 13: **end for**
 - 14: **return** YES
-

Algorithm 2 MHR Regularized Learner

Require: m i.i.d. samples from a distribution $\mathcal{D}$ and a distortion parameter α

Ensure: Myerson's optimal auction $M_{E'}$ with respect to E'

- 1: Let $E = \prod_{i=1}^n E_i$ be the empirical distribution over the samples
 - 2: **for** $i \leftarrow 1$ to n **do**
 - 3: Construct $\hat{E}_i$ as follows:
 - 4: Let $q_{E_i}(v)$ be the quantile function of E_i
 - 5: Define

$$c = \sqrt{\frac{2q_{E_i}(v)(1-q_{E_i}(v)) \ln(2mn\delta^{-1})}{m}} + \frac{4 \ln(2mn\delta^{-1})}{m} + \alpha$$
 - 6:
$$q_{\hat{E}_i}(v) = \begin{cases} \max(0, q_{E_i}(v) - c) & \text{if } v > 0, \\ 1 & \text{if } v = 0 \end{cases}$$
 - 7: $\tilde{E}_i = h_M^{-1}(\text{Conv}(h_M(\hat{E}_i(\cdot))))$
 - 8: Construct E'_i as follows:
 - 9:
$$q_{E'_i}(v) = \begin{cases} \max(0, q_{\tilde{E}_i}(v) - \alpha) & \text{if } v > 0, \\ 1 & \text{if } v = 0 \end{cases}$$
 - 10: **end for**
 - 11: Set $E' = \prod_{i=1}^n E'_i$
 - 12: **return** Myerson's optimal auction $M_{E'}$
-

4.2.3 Supporting Lemmas for MHR Distributions

We now establish key supporting lemmas for our MHR extension.

Lemma 4 (MHR Distributions are not Dense). *There exists a distribution D that is not MHR and an $\alpha > 0$ such that for all MHR distributions D' , it holds that $d_{KS}(D, D') > \alpha$.*

Proof. By Lemma 4 of the Safely Learning Optimal Auctions working paper, we know that regular distributions are not dense. That is, we know that there exists an irregular distribution D and an $\alpha > 0$ such that for all regular distributions D' , it holds that $d_{KS}(D, D') > \alpha$.

Because MHR implies regularity ($MHR \rightarrow Regular$), we know all irregular distributions are not MHR ($\neg Regular \rightarrow \neg MHR$). Thus, we know that there exists a non-MHR distribution D and an $\alpha > 0$ such that for all regular distributions D' , it holds that $d_{KS}(D, D') > \alpha$. Because this non-MHR distribution is more than α -far from all regular distributions, we also know that it is at least α -far from all MHR distributions. $\square$

Lemma 5 (Non-MHR Distributions are Dense). *For any distribution D and $\alpha > 0$, there exists a non-MHR distribution D' such that $D' \in B_{KS,\alpha}(D)$.*

Proof. In the previous Lemma, we established that irregular distributions are not MHR. Thus, we can carry over the findings from Lemma 5 of the Safely Learning Optimal Auctions working paper and say that For any distribution D and $\alpha > 0$, there exists a non-MHR distribution D' such that $D' \in B_{KS,\alpha}(D)$. $\square$

Lemma 6 (Preserving MHR under Probability Mass Addition). *Let D be an MHR distribution with CDF $F_D(x)$ and $\alpha \in (0, 1)$. Define the distribution D' with the following CDF:*

$$F_{D'}(x) = \begin{cases} F_D(x) + \alpha, & 0 \leq x < F_D^{-1}(1 - \alpha) \\ 1, & x \geq F_D^{-1}(1 - \alpha). \end{cases} \quad (3)$$

Then, D' is also an MHR distribution.

Proof. By Lemma 3, D' is MHR if and only if its link function $h_M(x; F_{D'})$ is convex. We will take the derivative of $h_M(x; F_{D'})$ and show that it is increasing.

For $0 \leq x < F_D^{-1}(1 - \alpha)$:

$$\begin{aligned} \frac{\partial h_M(x; F_{D'})}{\partial x} &= \frac{\partial}{\partial x}(-\ln(1 - F_{D'}(x))) \\ &= \frac{\partial}{\partial x}(-\ln(1 - (F_D(x) + \alpha))) \\ &= \frac{f_D(x)}{1 - F_D(x) - \alpha} \end{aligned}$$

For an MHR distribution D , $\frac{f_D(x)}{1-F_D(x)}$ is increasing. Since α is a constant, $\frac{f_D(x)}{1-F_D(x)-\alpha}$ is also increasing.

Therefore, $h_M(x; F_{D'})$ is convex, proving that D' is MHR. $\square$

4.2.4 Main Theorem for MHR Distributions

We now present the MHR extension of our main theorem, which provides a tighter revenue guarantee than the original result for regular distributions.

Theorem 7 (Testable Learning for MHR Distributions). *Let $D = \Pi_{i=1}^n D_i$ be a product distribution, let $\alpha \in (0, 1)$ be a distortion parameter, and let $\delta > 0$. Additionally, fix $m = \tilde{\Omega}(\log(1/\delta)/\alpha^2)$. Suppose $\mathcal{T}$ is the tester described in Algorithm 1 (with MHR link function) and $\mathcal{L}$ is the learning algorithm described in Algorithm 2 (with MHR link function). Then, $\mathcal{T}$ is an m -sample regularized tester for the $(m, n\alpha)$ -regularized learner $\mathcal{L}$.*

Proof. We begin by showing that the tester guarantee holds with probability at least $1 - \delta$. Choose m large enough such that:

$$\sqrt{\frac{2q_{D_i}(v)(1 - q_{D_i}(v)) \ln(2mn\delta^{-1})}{m}} + \frac{4 \ln(2mn\delta^{-1})}{m} \leq \alpha$$

for every $i \in [n]$.

Suppose there exists an MHR product distribution D' such that $D'_i \in B_{KS,\alpha}(D_i)$ for all $i \in [n]$. By Lemma 5 of Guo et al. (2019), with probability $1 - \delta$, for every value v and every $i \in [n]$, it holds that:

$$|q_{E_i}(v) - q_{D_i}(v)| \leq \sqrt{\frac{2q_{D_i}(v)(1 - q_{D_i}(v)) \ln(2mn\delta^{-1})}{m}} + \frac{4 \ln(2mn\delta^{-1})}{m} \leq \alpha.$$

Fix an index $i \in [n]$. To show the correctness of the tester, we will prove that $|q_{\tilde{E}_i} - q_{E_i}| \leq 2\alpha$. Observe that by construction, $D'_i \succeq \tilde{E}_i \succeq \hat{E}_i$. By a sandwiching argument, we only need

to upper bound the quantile distance between D'_i and E_i and between $\hat{E}_i$ and E_i , respectively:

$$|q_{D'_i}(v) - q_{E_i}(v)| \leq |q_{D'_i}(v) - q_{D_i}(v)| + |q_{D_i}(v) - q_{E_i}(v)| \leq 2\alpha$$

and

$$|q_{\hat{E}_i}(v) - q_{E_i}(v)| \leq 2\alpha.$$

Thus, the tester outputs YES with probability at least $1 - \delta$.

We now prove the learner guarantee. Suppose $\mathcal{T}(D, \alpha, m)$ outputs YES and let M be the mechanism outputted by $\mathcal{L}(D, \alpha, m)$ (i.e., $\text{Rev}(M, E') = \text{OPT}(E')$). As before, we note that $D'_i \succeq \tilde{E}_i \succeq E'_i$ and that $D_i \succeq E'_i$ by construction. Additionally, by Lemma 6, E' is MHR.

We need to bound the distance between E'_i and any MHR distribution $D'_i \in B_{KS, \alpha}(D_i)$ for all $i \in [n]$. Using the triangle inequality:

$$\begin{aligned} |q_{E'_i}(v) - q_{D'_i}(v)| &\leq |q_{E'_i}(v) - q_{\tilde{E}_i}(v)| + |q_{\tilde{E}_i}(v) - q_{E_i}(v)| + |q_{E_i}(v) - q_{D_i}(v)| + |q_{D_i}(v) - q_{D'_i}(v)| \\ &\leq \alpha + 2\alpha + \alpha + \alpha = 5\alpha. \end{aligned}$$

Using this bound, we can now apply Theorem 4.4 of Guo et al. (2021) to show that with probability at least $1 - \delta$, it holds that

$$\text{OPT}(E') \geq (1 - O(n\alpha))\text{OPT}(D')$$

for any MHR product distribution $D' = \prod_{i=1}^n D'_i$ such that for all $i \in [n]$ it holds that $D'_i \in B_{KS, \alpha}(D_i)$. To complete the proof, we use that $D \succeq E'$ and apply strong revenue monotonicity from Theorem 2 of the working paper to conclude that

$$\text{Rev}(M, D) \geq \text{OPT}(E') \geq (1 - O(n\alpha))\text{OPT}(D')$$

The key improvement from regular distributions is that we have replaced $O(\sqrt{n\alpha})$ with $O(n\alpha)$ in the bound, removing the square root and providing a tighter guarantee for MHR distributions. $\square$

4.2.5 Comparison with Regular Distribution Bounds

To highlight the improvement of our MHR extension, we compare the revenue guarantees for both distribution families:

- **Regular Distributions:** $Rev(M, D) \geq (1 - O(\sqrt{n\alpha}))OPT(D')$
- **MHR Distributions:** $Rev(M, D) \geq (1 - O(n\alpha))OPT(D')$

When α is small (which is the typical case in practice), the MHR bound provides a significantly tighter guarantee. For example, if $\alpha = 0.01$ and $n = 1$:

- Regular bound: $Rev(M, D) \geq (1 - O(0.1))OPT(D')$
- MHR bound: $Rev(M, D) \geq (1 - O(0.01))OPT(D')$

4.2.6 MHR Extensions for Additional Results

Our MHR extension naturally carries forward to other results in the paper. Specifically, the MHR variants of Theorems 8 and 10 would similarly replace the $O(\sqrt{n\alpha})$ term with $O(n\alpha)$ based on applying Theorem 4.4 (instead of 4.3) of Guo et al. (2021), yielding tighter guarantees when testing the Bulow-Klemperer Theorem and the Anonymous Price Auction for MHR distributions. Formally, this changes the results into the following:

Bulow-Klemperer:

- **Regular Distributions:** $Rev_{n+1}(VA, D) \geq (1 - O(\sqrt{na})) \cdot OPT_n(D')$
- **MHR Distributions:** $Rev_{n+1}(VA, D) \geq (1 - O(na)) \cdot OPT_n(D')$

Anonymous Price Auction:

- **Regular Distributions:** $Rev(APA(p^*), \mathbf{D}) \geq \frac{1}{e}(1 - O(\sqrt{na}))OPT(\mathbf{D}')$
- **MHR Distributions:** $Rev(APA(p^*), \mathbf{D}) \geq \frac{1}{e}(1 - O(na))OPT(\mathbf{D}')$

4.3 Analysis on Improving the Upper Bounds

Our analysis also explored the issue on if we could further tighten the sample complexity bounds provided in Guo et al. (2019). Guo et al. provide tight bounds for the sample complexity by offering the lower bound $\Omega(n\varepsilon^{-3})$ and the upper bound $\tilde{O}(n\varepsilon^{-3})$ for regular distributions (and similar bounds for MHR distributions).

The tilde in big O notation implies that there are hidden logarithmic factors within the upper bound. Specifically, for regular distributions, Guo et al. found that the upper bound was

$$O(n\varepsilon^{-3} \ln(n\varepsilon^{-1}) \ln(n\delta^{-1}\varepsilon^{-1})) = \tilde{O}(n\varepsilon^{-3})$$

We conducted a rigorous analysis of this paper to identify

- Where each logarithmic term was coming from and
- If there was a way to tighten (or eliminate) some of these factors.

4.3.1 Sources of the Logarithmic Factors

Our analysis identified two distinct logarithmic factors in the sample complexity bounds:

The First Logarithmic Factor: $\ln(2mn\delta^{-1})$

This factor appears in the definition of the function $s_{m,n,\delta}(q)$ used to construct the “dominated empirical distribution” $\tilde{E}$:

$$s_{m,n,\delta}(q) = \max \left(0, q - \sqrt{\frac{2q(1-q) \ln(2mn\delta^{-1})}{m}} - \frac{4 \ln(2mn\delta^{-1})}{m} \right)$$

This factor emerges from applying Bernstein’s inequality in Lemma 5, which bounds the difference between the empirical and true quantiles. The authors apply a union bound over all m possible sample values and all n bidders, resulting in a failure probability of at most $\frac{\delta}{mn}$ for each value-bidder pair. The combined probability of any failure is at most δ , which introduces the $\ln(2mn\delta^{-1})$ term.

The factor represents the statistical price paid for ensuring the bounds hold simultaneously for all bidders and all possible values with high probability $(1-\delta)$. This is a fundamental requirement for the algorithm's correctness guarantee.

The Second Logarithmic Factor: $\ln(n\varepsilon^{-1})$

This factor emerges from the information-theoretic analysis used to bound the difference between the optimal revenues of the true distribution D and the doubly-shaded distribution $\tilde{D}$. Unlike the first logarithmic factor (which comes from concentration bounds), this factor is inherent to the specific construction needed to prove revenue guarantees for regular distributions.

For regular distributions, Section 3.3 (page 19) of Guo et al. (2019) defines these key parameters:

- $p = \varepsilon/8$
- Upper bound thresholds: $\bar{v}_i^2 = \sup\{v : q^{D_i}(v) \geq p\varepsilon^2/n\} = \sup\{v : q^{D_i}(v) \geq \varepsilon^3/(8n)\}$

These choices create:

1. A point mass at the lower endpoint with probability $p_\ell = \varepsilon$ (from the bottom truncation)
2. A point mass at the upper endpoint with probability $p_u = \varepsilon^3/(8n)$ (from the construction of $\bar{v}_i^2$)

In Lemma 10 (pages 12-13), the KL divergence between distributions D' and $\tilde{D}'$ depends on:

$$\ln(p_\ell^{-1}p_u^{-1}) = \ln(\varepsilon^{-1} \cdot 8n/\varepsilon^3) = \ln(8) + \ln(n) + 4\ln(\varepsilon^{-1})$$

This expression also appears in the bound:

$$D_{SKL}(D', \tilde{D}') = O\left(\frac{\ln(mn\delta^{-1})}{m} \ln(p_\ell^{-1}p_u^{-1})\right)$$

In asymptotic notation, the constant $\ln(8)$ is absorbed into the big-O, and $4\ln(\varepsilon^{-1})$ simplifies to $O(\ln(\varepsilon^{-1}))$. Putting this all together, we see the $O(\ln(n\varepsilon^{-1}))$ term arising.

This logarithmic factor represents the “cost” of constructing surrogate distributions that simultaneously:

1. Have bounded support (for concentration bounds to work)
2. Remain close to the original distribution (for revenue guarantees)
3. Preserve the MHR/regular property (for applying Myerson’s characterization)

Thus, it seems as though the $\ln(n\varepsilon^{-1})$ factor is fundamental to the information-theoretic approach and likely cannot be removed without developing an entirely different analytical technique.

4.3.2 Areas of Possible Improvement

After careful analysis, we identified several potential avenues for improving these logarithmic factors that may be used as further research:

1. Strengthened DKW Inequality with Quantile-Dependent Bounds

As noted by Guo et al., a strengthened version of the Dvoretzky-Kiefer-Wolfowitz (DKW) inequality with quantile-dependent bounds could reduce the logarithmic overhead. Standard DKW provides a uniform bound on CDF estimation error without requiring a union bound over values, but doesn’t provide the quantile-dependent precision that their analysis leverages.

In our research, we conducted an exercise that hypothesized how the results would change if a version of the DKW inequality and Bernstein’s inequality without the $\ln(2mn\delta^{-1})$ factor, and we found that this logarithmic factor would no longer propagate through the proof, resulting in a tighter final bound.

Development of a hybrid approach that combines the uniformity of DKW with the precision of quantile-dependent bounds could potentially eliminate or reduce the first logarithmic

factor.

2. Applying Theory of Classification

Boucheron (2010) offers interesting research into improving Bernstein’s inequality through empirical risk minimization that may have some applications to tighten the sample complexity bounds found by Guo et al. Their work on data-dependent concentration inequalities could provide sharper bounds that adapt to the actual observed samples rather than worst-case scenarios. Additionally, the paper’s exploration of localized Rademacher complexities might offer a path to developing more refined bounds in the auction setting, particularly for distribution classes with structured tail behavior like MHR distributions. Finally, their techniques for model aggregation could inspire new approaches that combine multiple auction mechanisms to achieve better sample complexity with only logarithmic overhead, potentially eliminating one of the logarithmic factors in Guo et al.’s bounds.

5 Conclusion

The goals of this thesis were to extend the Testable Learning Framework for mechanism design to MHR distributions, and demonstrate both theoretical and practical improvements over existing approaches for regular distributions.

We established tighter revenue guarantees for MHR distributions ($1 - O(n\alpha)$) versus ($1 - O(\sqrt{n\alpha})$) for regular distributions representing a significant improvement, especially when the distortion parameter α is small. Our empirical testing validated the theoretical predictions, showing that the tester behaves largely as expected, accepting when $\alpha \geq \varepsilon$ and rejecting otherwise.

Our implementation and testing of the regularized tester-learner pair revealed several key insights. First, sample size matters significantly—with sufficient samples, the tester’s behavior closely matches theoretical expectations. Second, the learned revenue approaches the theoretical optimal as α decreases, with the curve following a similar slope to the theoretical

bound. Third, the framework shows surprising robustness in handling mixed distributions that exhibit varying degrees of irregularity.

The analysis of logarithmic factors in the sample complexity bounds identified potential areas for improvement, including strengthened concentration inequalities and application of classification theory. These refinements could further reduce the sample requirements for learning near-optimal auctions.

The testable learning approach represents a significant advance in practical auction design by bridging the gap between theoretical guarantees and real-world applications where distributional assumptions may not hold. By extending this framework to MHR distributions, we’ve demonstrated that stronger structural assumptions can yield tighter guarantees without sacrificing the framework’s applicability.

Future work could explore extensions to multi-bidder and multi-item settings, test performance on additional distribution families, and implement the suggested improvements to reduce logarithmic factors in the sample complexity bounds.

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